Technische Universität Braunschweig

DLR

An Image Semantic Segmentation Model of CFRP Micrographs

Team 10

Narunat Pantapalin and Vinit Firke

1. Introduction

This project aims to develop an image semantic segmentation model capable of classifying different regions within CFRP micrographs into four distinct categories, which can be used to detect anormalities in CFRP mircrographs.

- 0° Fiber
- 90° Fiber
- Resin
- Other

The deep learning models are trained using high-resolution microscopic images of CFRP to automate the segmentation process, facilitate defect identification and material analysis.

2. Methodology

Preprocessing

- The dataset consists of **high-resolution microscopic images**.
- Images were divided into subimages of size **512 x 512** for training.

Model Architectures: U-Net Architectures

- Efficient for learning spatial hierarchies and captures both global context and fine-grained details.
- Suitable for small dataset

2 Variants:

- **U-Net:** Standard Unet.
- **Squeeze and Excitation U-Net:** Standard Unet with squeeze and excitation block in the bottleneck step to improve feature representation.

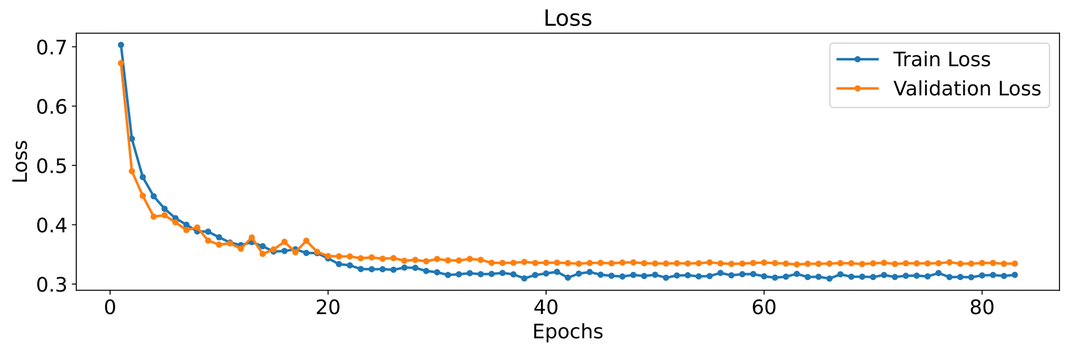
3. Training and Evaluation Process

Training:

- **Objective function**: Mean IOU
- **Optimizer**: ADAM with 10^{-4} Learning rate, reduce on plateau with 0.1 factor and early stopping for improving generalization

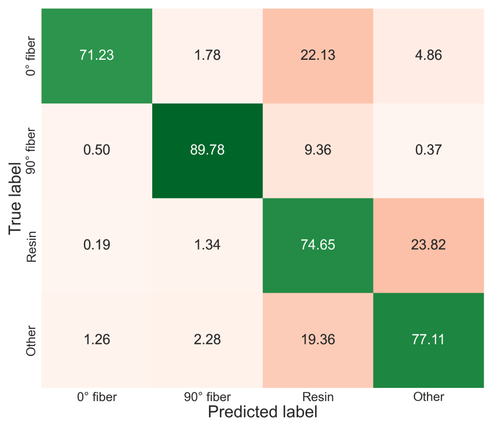
Evaluation: Performance was assessed using key metrics: mean IOU, dice coefficient and pixel accuracy

4. Training Progress



5. Results and Discussion

Model	Pixel Accuracy	Mean IOU
U-Net	67.75%	43.44%
Squeeze and Excitation U-Net	79.96%	56.19%



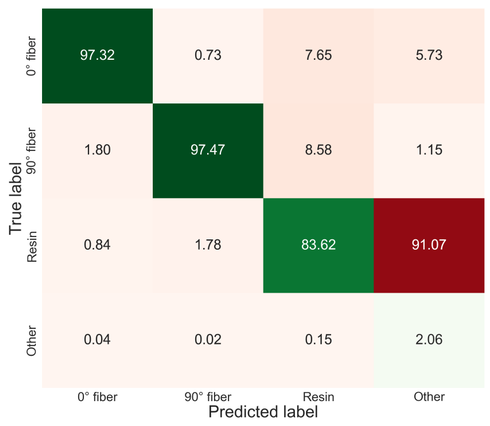


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Image Segmentation Challenge

Final Presentation on 12.02.2025